FORM TP 2007248

TEST CODE **02134032**

MAY/JUNE 2007

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

PURE MATHEMATICS

UNIT 1 – PAPER 03/2

$1\frac{1}{2}$ hours

18 MAY 2007 (p.m.)

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This examination paper consists of THREE sections: Module 1, Module 2, and Module 3.

Each section consists of 1 question.
The maximum mark for each section is 20.
The maximum mark for this examination is 60.
This examination paper consists of 4 pages.

INSTRUCTIONS TO CANDIDATES

1. **DO NOT** open this examination paper until instructed to do so.
2. Answer **ALL** questions from the **THREE** sections.
3. Unless otherwise stated in the question, any numerical answer that is not exact **MUST** be written correct to three significant figures.

Examination materials

Mathematical formulae and tables
Electronic calculator
Graph paper

Section A (Module 1)

Answer this question.

1. (a) Given that $h(x) = x + \frac{1}{x}$, $x \neq 0$, express in terms of t,

(i) $h(t)h\left(\frac{1}{t}\right)$ [3 marks]

(ii) $h(t)h\left(\frac{1}{t}\right) - h\left(t^2\right)$. [4 marks]

(b) Solve $\frac{x+1}{x+2} > 2$, $x \neq -2$, $x \in \mathbf{R}$. [6 marks]

(c) The function, h, is defined on $\mathbf{R}$ by

$h: x \rightarrow x^2 - 2x + 3$, $x \geq 1$.

(i) Determine the range of h. [2 marks]

(ii) Explain why the inverse function h^{-1} of h exists. [3 marks]

(iii) Determine the domain of h^{-1}. [2 marks]

Total 20 marks

Section B (Module 2)

Answer this question.

2. The parametric equations of an ellipse are given as

$$x = 3\cos\theta$$
$$y = 4\sin\theta.$$

(a) Show that the Cartesian equation of the ellipse is $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1.$ **[3 marks]**

(b) Find the equation of the tangent at the point P, where $\theta = \dfrac{\pi}{6}$. **[5 marks]**

(c) The tangent at P cuts the x and y axes (origin at O) at Q and R respectively.

Determine

(i) the coordinates of Q and R **[4 marks]**

(ii) the area of ΔQOR **[3 marks]**

(iii) the length of QR **[3 marks]**

(iv) the length of the perpendicular from O to QR. **[2 marks]**

Total 20 marks

Section C (Module 3)

Answer this question.

3. (a) (i) Given that $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$, factorize $x - 1$ in terms of $x^{1/3}$. [2 marks]

(ii) Hence, or otherwise, evaluate $\lim_{x \to 1} \dfrac{x^{2/3} - 1}{x^2 - 1}$. [4 marks]

(b) (i) Find the critical values of the function $y = x^3 - x^2 - x + 1$. [5 marks]

(ii) Classify, giving reasoned arguments, the critical values of y. [3 marks]

(c) A chemical factory wishes to make a **closed** cylindrical container of thin metal to hold a volume V of 10 cm^3. The outside surface of the container is S cm^2, the radius is r cm and the height is h cm.

(i) Show that $S = 2\pi r^2 + \dfrac{20}{r}$. [2 marks]

(ii) Hence, show that S has a minimum value when $r^3 = \dfrac{5}{\pi}$.

[$V = \pi r^2 h$, $S = 2\pi r^2 + 2\pi rh$] [4 marks]

Total 20 marks

END OF TEST

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